PARISHRAM 2.0

CLASS 12TH 2026

Biology-

HUMAN REPRODUCTION

ZOOLOGY | LECTURE-1

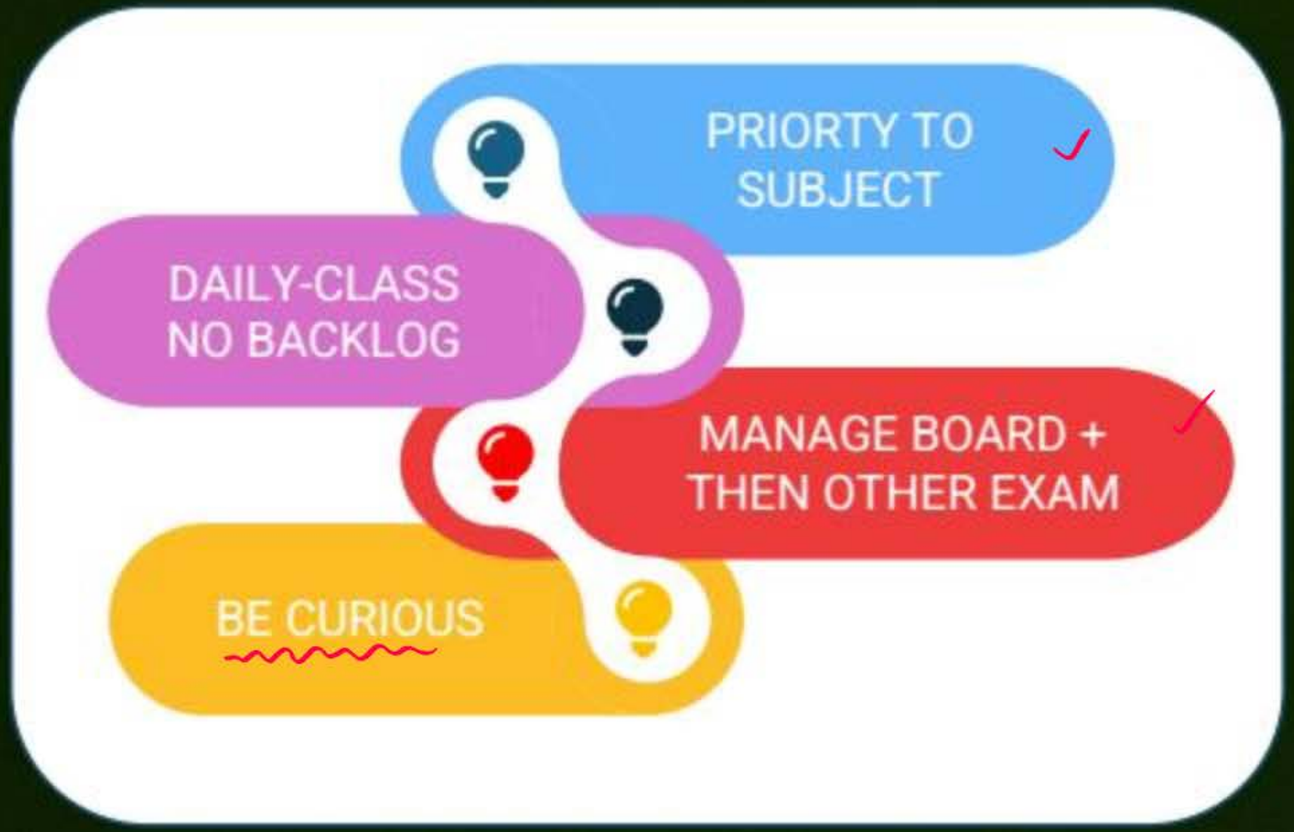
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TODAY'S TOPIC –

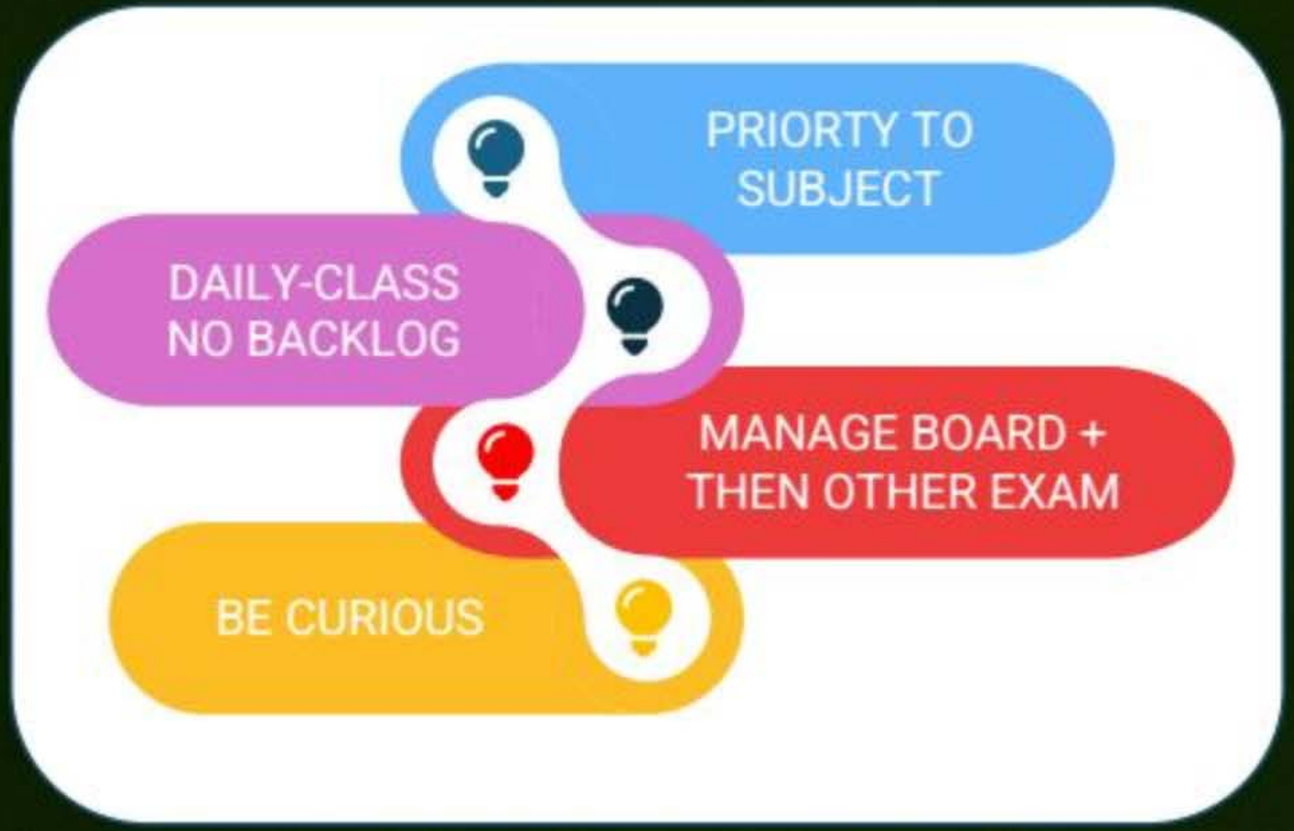
1. INTRODUCTION – REPRODUCTION
2. TYPE OF SEX ORGAN IN MALE
3. MALE REPRODUCTIVE SYSTEM

DEMAND



REQUEST

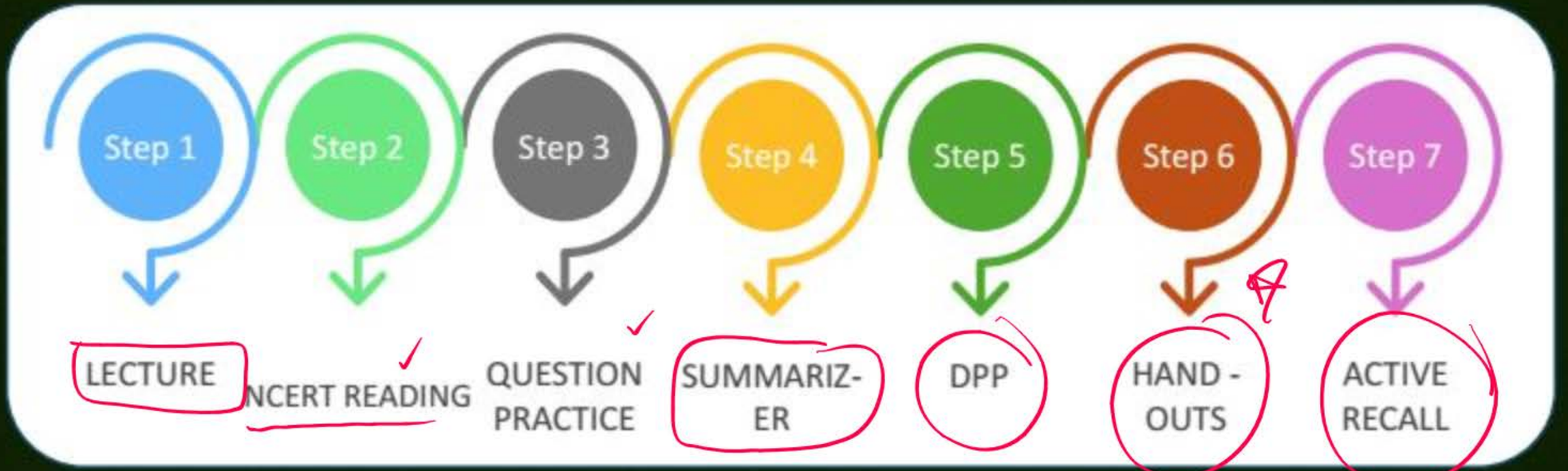
DEMAND



REQUEST



BATCH FLOW


Step 1
Step 2
Step 3
Step 4
Step 5
Step 6
Step 7
LECTURE
NCERT READING
QUESTION PRACTICE
SUMMARIZ-ER
DPP
HAND - OUTS
ACTIVE RECALL

REPRODUCTION

Biological process to produce offspring
Progeny
Baby

Reproduction
- Asexual
- Sexual *

Animal
- * Viviparous → Young baby
- OVIPAROUS

REPRODUCTION



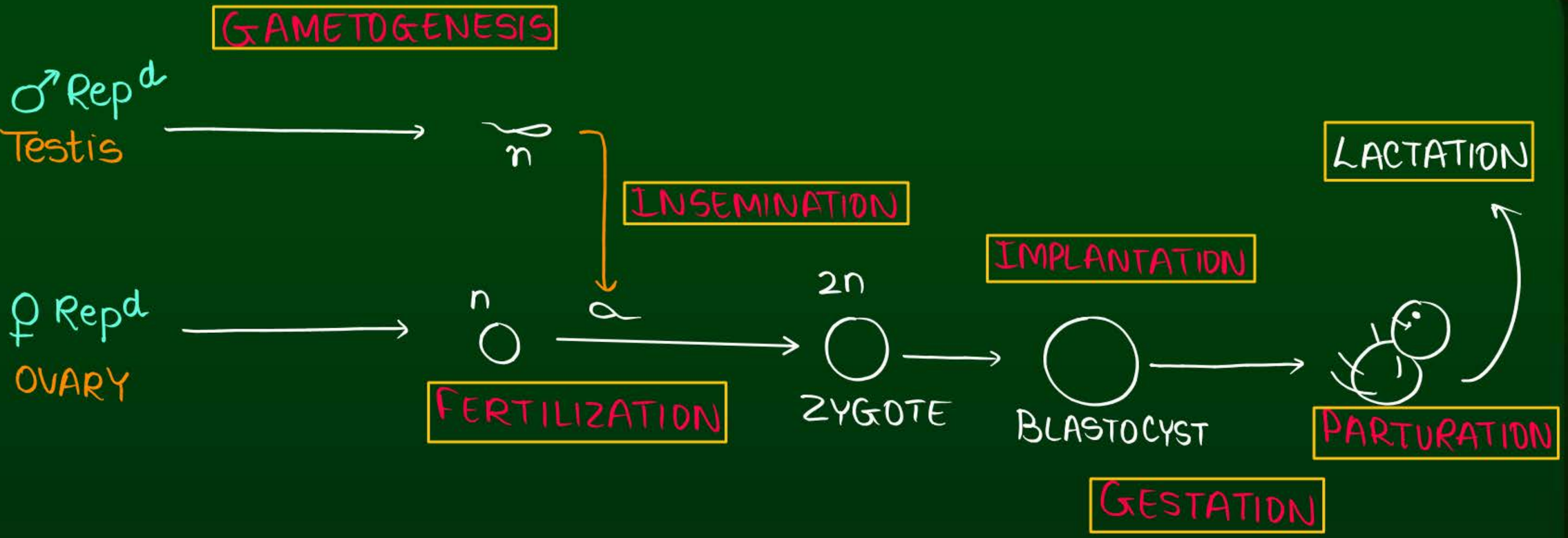
Biological process to produce offspring for continuity of species over generation

Girl

Boy

- Sexually reproducing
- Viviparous
- Sexual dimorphism
- Reproduction start at puberty

OVERVIEW OF EMBRYONIC DEVELOPMENT



NCERT

As you are aware, humans are sexually reproducing and viviparous. The reproductive events in humans include formation of gametes (gametogenesis), i.e., sperms in males and ovum in females, transfer of sperms into the female genital tract (insemination) and fusion of male and female gametes (fertilisation) leading to formation of zygote. This is followed by formation and development of blastocyst and its attachment to the uterine wall (implantation), embryonic development (gestation) and delivery of the baby (parturition). You have learnt that these reproductive events occur after puberty. There are remarkable differences between the reproductive events in the male and in the female, for example, sperm formation continues even in old men, but formation of ovum ceases in women around the age of fifty years. Let us examine the male and female reproductive systems in human.

EXTRACT

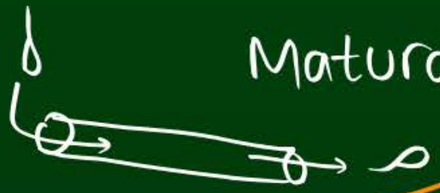
→ Sperm formation continue in male even in old age

→ While infemal cease at age of 50.

TYPE OF SEX ORGAN IN MALE

1° sex organ
TESTIS
- Formation of Gamete
- Release hormone

2° sex organ
Maturation and transfer of gamete



DUCT
→ Rete testis
→ Vasa efferentia
→ Epididymus
→ Vas deferens
→ Ejaculatory duct
→ Urethra

Gland
→ Seminal vesicle
→ Prostate
→ Bulbourethal or Cowper's gland

External genitalia
- Penis
- Scortum

NCERT

EXTRACT

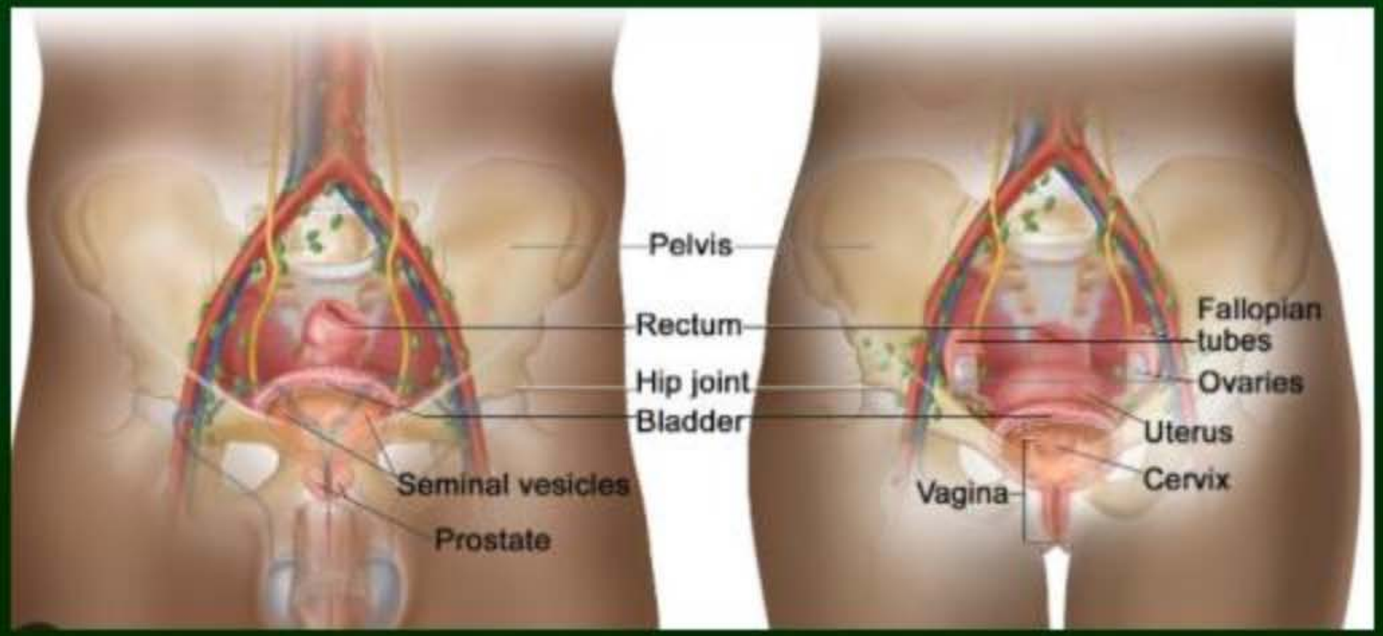
2.1 The Male Reproductive System

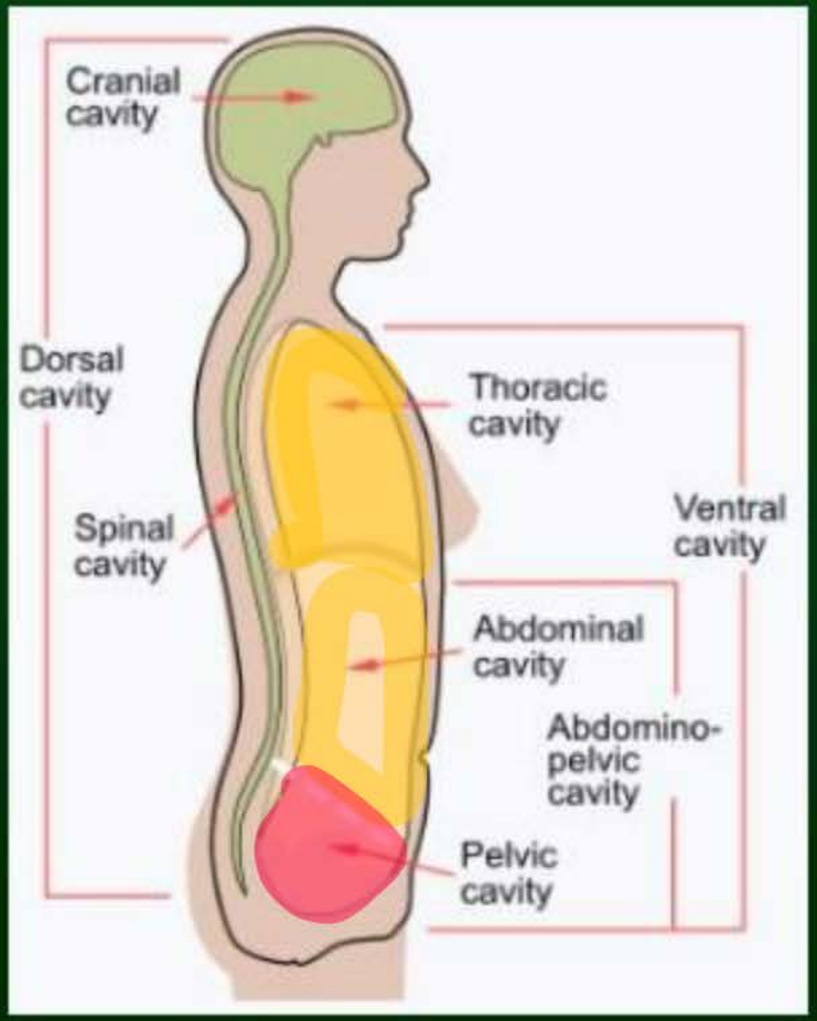
The male reproductive system is located in the pelvis region (Figure 2.1a). It includes a pair of **testes** alongwith **accessory ducts**, **glands** and the **external genitalia**.

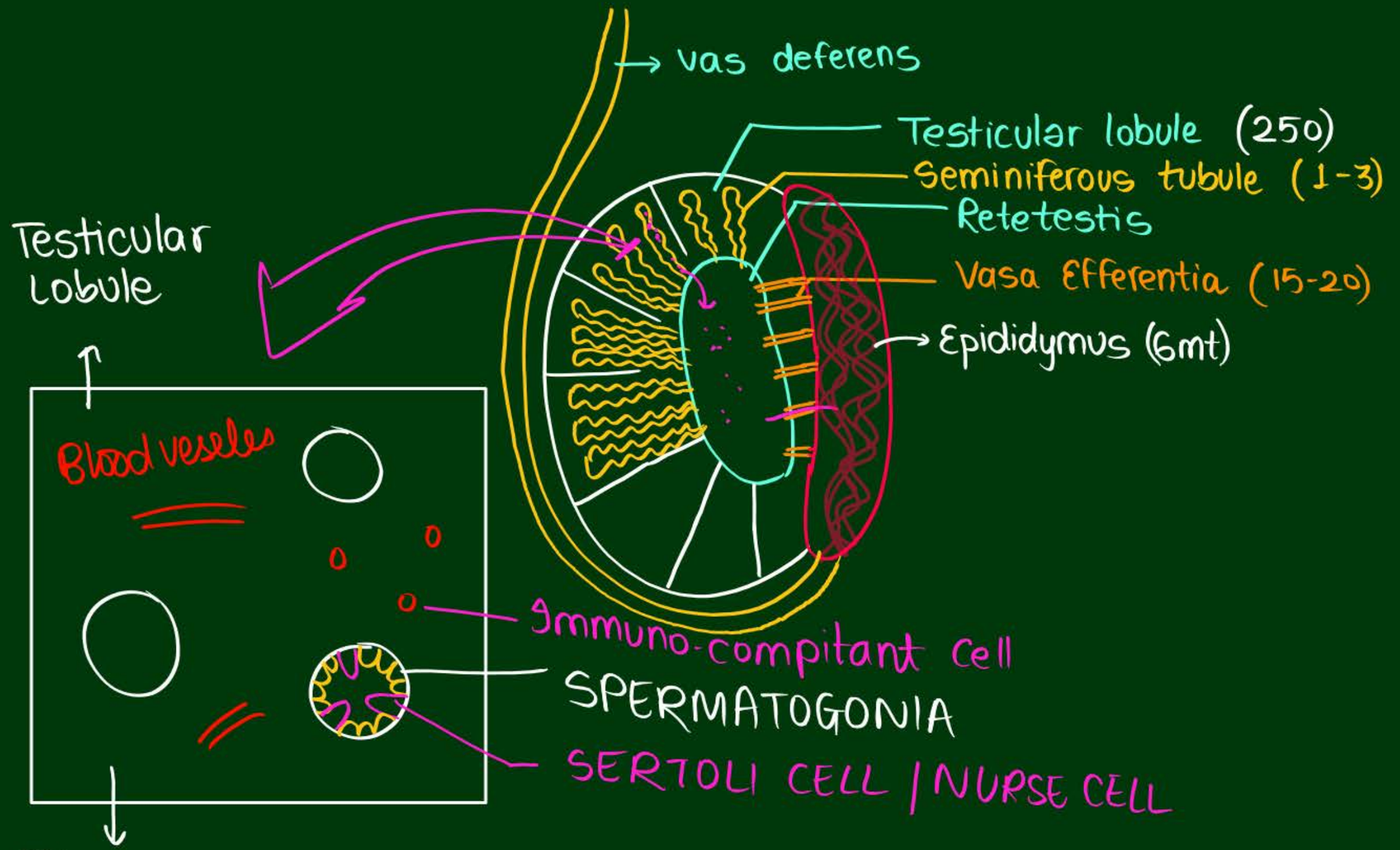
NCERT

2.1 The Male Reproductive System

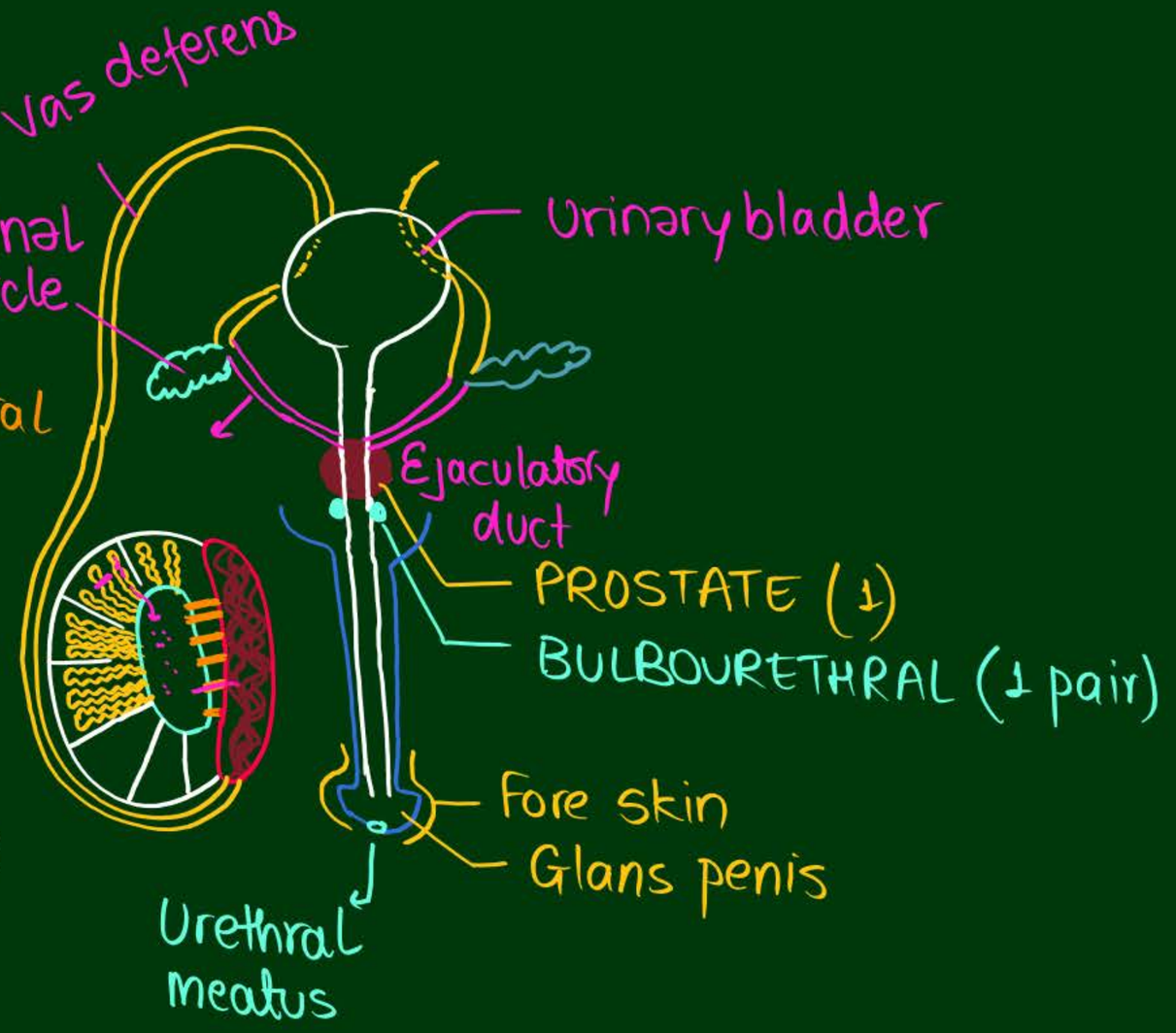
The male reproductive system is located in the pelvis region (Figure 2.1a). It includes a pair of **testes** alongwith **accessory ducts, glands** and the **external genitalia**.







Testicular lobule (250)
↓
Seminiferous tubule 1-3
↓
Rete testis
↓
Vasa Efferentia (15-20)
↓
Epididymus (6mt)
↓
Vas deferens



	S.V	Prostate	bulbourethral
Location	Posterior to urinary bladder	below Urinary bladder	below Prostate
amount of Secretion	60-70%	25%-30%	5-10%

Seminal plasma

- fructose
- Prostaglandin
- Calcium
- Enzyme
- mucus

THE MALE REPRODUCTIVE SYSTEM

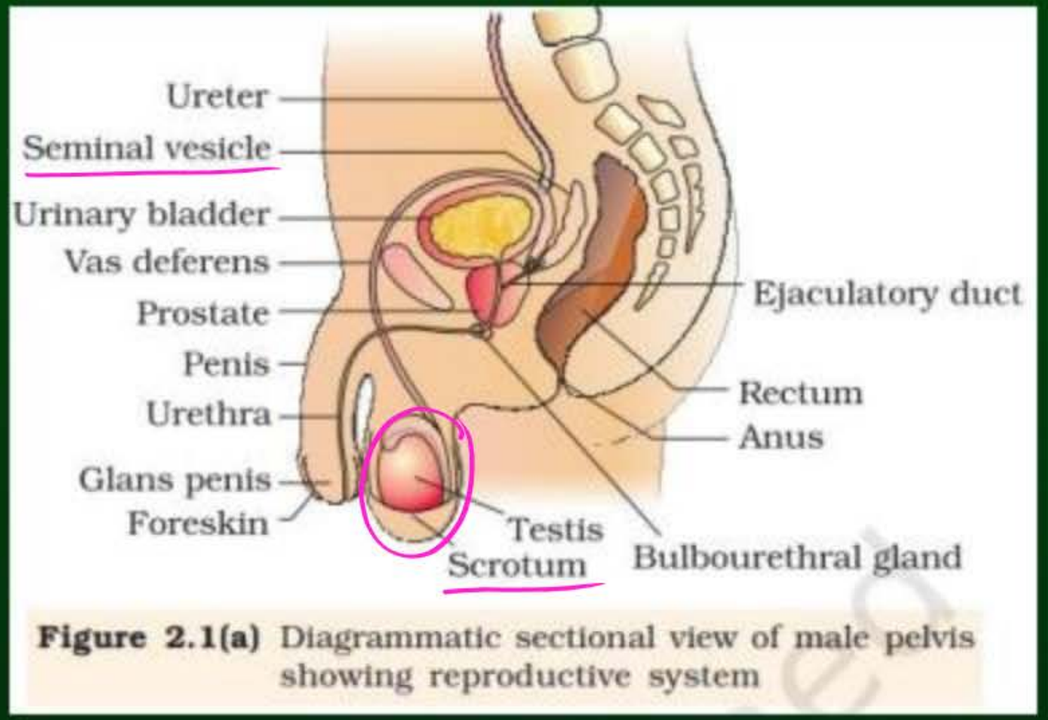
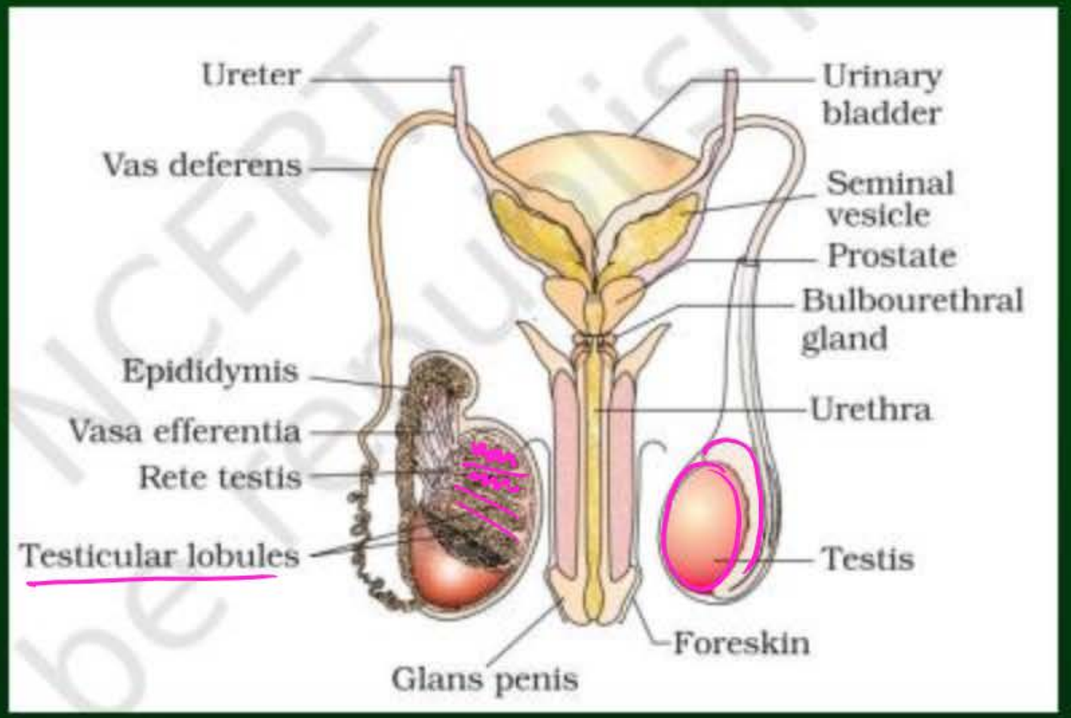


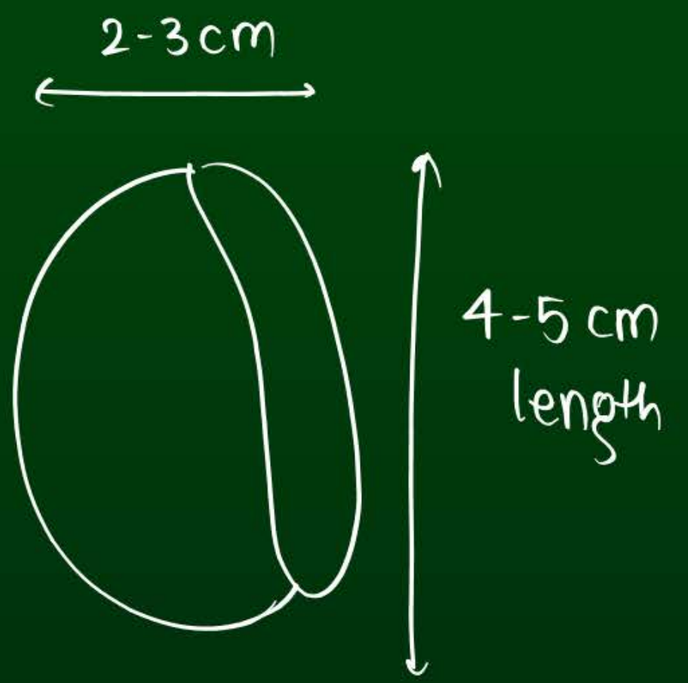
Figure 2.1(a) Diagrammatic sectional view of male pelvis showing reproductive system



NCERT

The testes are situated outside the abdominal cavity within a pouch called **scrotum**. The scrotum helps in maintaining the low temperature of the testes (2–2.5° C lower than the normal internal body temperature) necessary for spermatogenesis. In adults, each testis is oval in shape, with a length of about 4 to 5 cm and a width of about 2 to 3 cm. The testis is covered by a dense covering. Each testis has about 250 compartments called **testicular lobules** (Figure 2.1b).

EXTRACT



NCERT

Each lobule contains one to three highly coiled **seminiferous tubules** in which sperms are produced. Each seminiferous tubule is lined on its inside by two types of cells called **male germ cells** (*spermatogonia*) and **Sertoli cells** (Figure 2.2). The male germ cells undergo meiotic divisions finally leading to sperm formation, while Sertoli cells provide nutrition to the germ cells. The regions outside the seminiferous tubules called interstitial spaces, contain small blood vessels and **interstitial cells** or **Leydig cells** (Figure 2.2). Leydig cells synthesise and secrete testicular hormones called androgens. Other immunologically competent cells are also present.

EXTRAC

NCERT

EXTRACT

The male sex accessory ducts include **rete testis**, **vasa efferentia**, **epididymis** and **vas deferens** (Figure 2.1b). The seminiferous tubules of the testis open into the vasa efferentia through rete testis. The vasa efferentia leave the testis and open into epididymis located along the posterior surface of each testis. The epididymis leads to vas deferens that ascends to the abdomen and loops over the urinary bladder. It receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct (Figure 2.1a). These ducts store and transport the sperms from the testis to the outside through urethra. The urethra originates from the urinary bladder and extends through the penis to its external opening called **urethral meatus**.

The penis is the male external genitalia (Figure 2.1a, b). It is made up of special tissue that helps in erection of the penis to facilitate insemination. The enlarged end of penis called the glans penis is covered by a loose fold of skin called **foreskin**.

The male accessory glands (Figure 2.1a, b) include paired **seminal vesicles**, a **prostate** and paired **bulbourethral** glands. Secretions of these glands constitute the seminal plasma which is rich in fructose, calcium and certain enzymes. The secretions of bulbourethral glands also helps in the lubrication of the penis.

QUESTION

The testes are situated outside the abdominal cavity within a pouch called

1. Urethra
2. Scrotum ✓
3. Urinary bladder
4. Pelvic region



QUESTION

The regions outside the seminiferous tubules that contain Leydig cells are called

1. Interstitial space
2. Spermatogonium
3. Sertoli cells
4. Leydig cells



QUESTION

Which of the following is not a part of male reproductive system

1. Rete testis
2. Isthmus
3. Vasa efferentia
4. Epididymis



QUESTION

Which of the following represent the correct pathway of transport of sperms?

1. Rete testis → Vasa efferentia → Epididymis → Vas deference
2. Rete testis → Epididymis → Vasa efferentia → Vas deference
3. Rete testis → Vas deference → vasa efferentia → Epididymis
4. Vasa efferentia → Rete testis → Vas deference → Epididymis



QUESTION

Among the following which one is an accessory duct of male reproductive system?

1. Cervix
2. Uterus
3. Vas deferens
4. Fallopian tube



SUMMARIZER

Reproduction
↓
type of male sex organ
↓
DIAGRAM: Male repd sys

- 1°
- 2°
  - Duct
  - Gland
  - external genitalia

Homework

→ 2 Q

→ DPP ✓
10 Q

→

THANK YOU